

Car Value Drivers: Regression Analysis

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Data Analysis Project

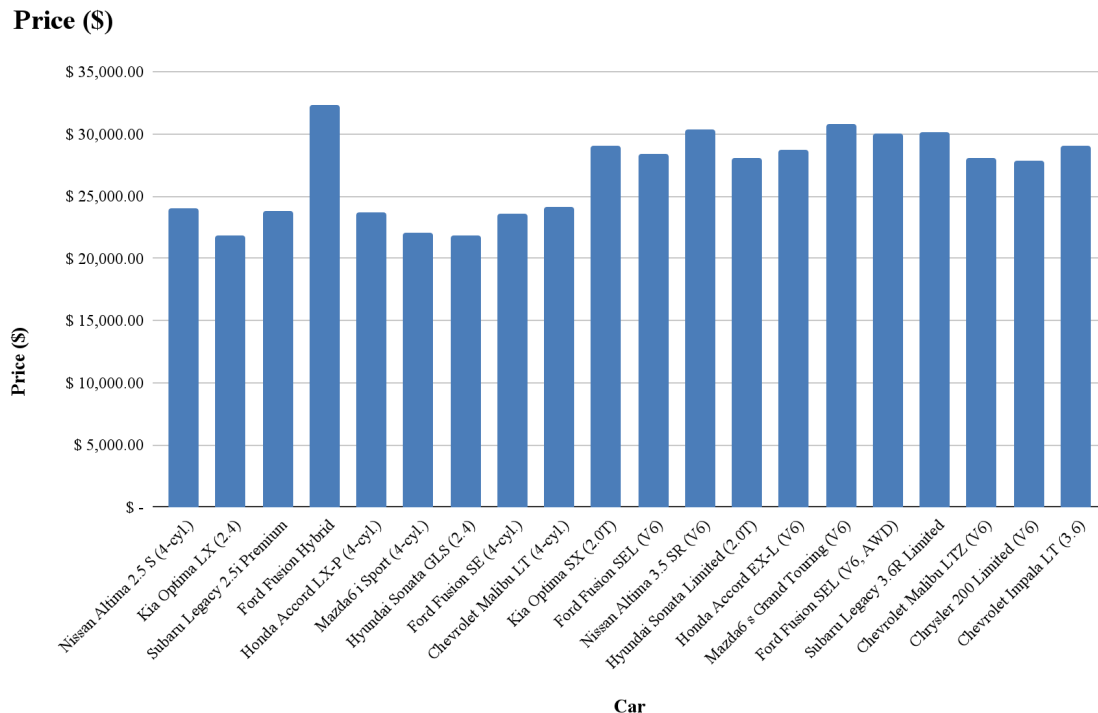
Microsoft Excel | Statistical Analysis

Summary Statistics

When developing numerical summaries of the data derived from the value score table for 20 family sedans, we can compute measures of center and variation for each quantitative component to describe its central tendency and spread.

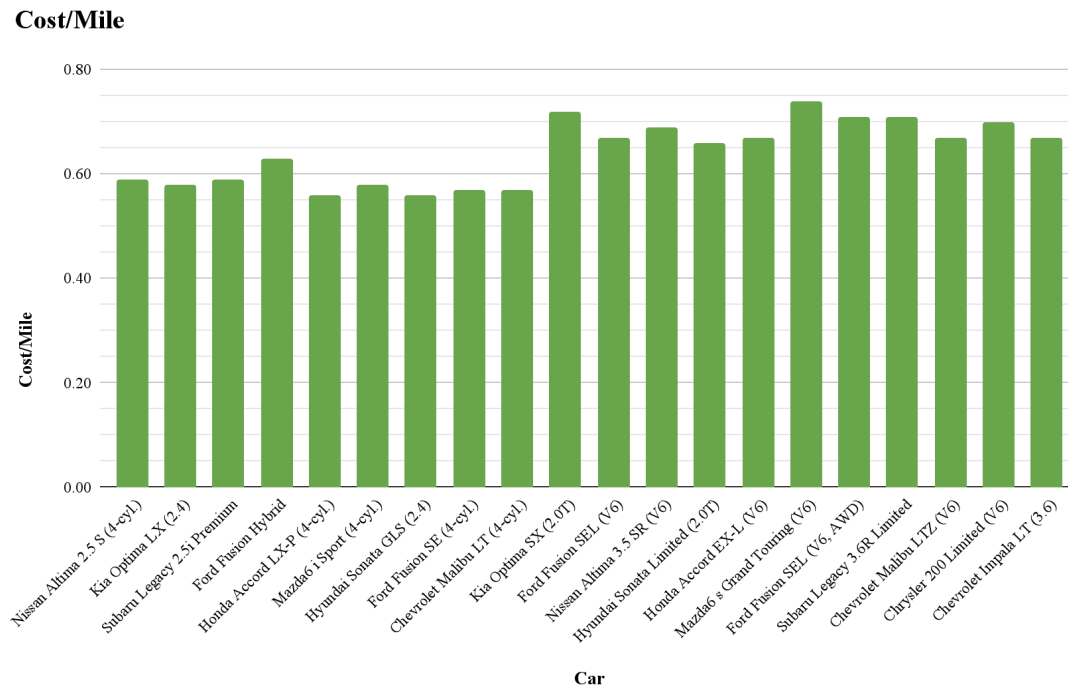
	Price (\$)	Cost/Mile	Road-Test Score	Value Score
Mean:	\$ 26,886.20	0.64	80.45	1.46
Median:	\$ 28,067.50	0.67	82.00	1.43
Standard deviation:	\$ 3,374.28	0.0612	9.9020	0.1974
Minimum:	\$ 21,800.00	0.56	52.00	1.05
Maximum:	\$ 32,360.00	0.74	93.00	1.75
Range:	\$ 10,560.00	0.18	41.00	0.70

These statistics provide an overview of how prices, operating costs, performance, and value scores vary among family sedans, helping identify patterns that will be important for later regression analysis. We can also visualize the data in graphs to compare each car's statistics.



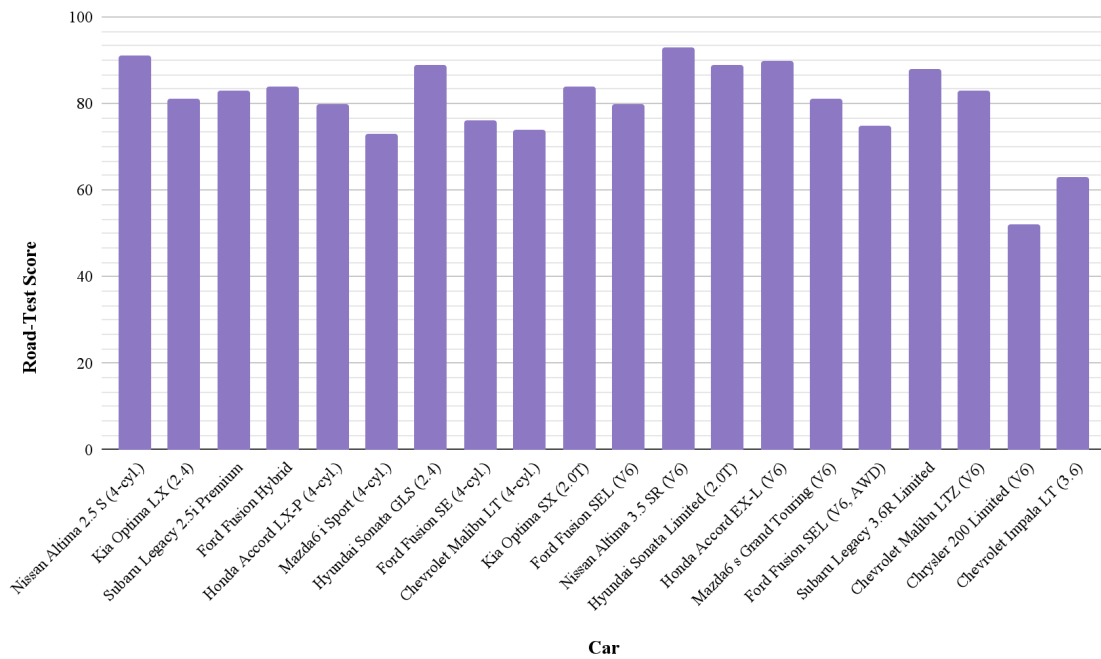
A histogram of the prices for the 20 family sedans shows how the vehicle costs are distributed across the dataset. Most family sedans fall within the mid-\$20,000 to low-\$30,000 range, indicating that the market for this vehicle class is relatively consistent in price. The distribution

does not appear extremely skewed, suggesting that the sedan prices in the sample are relatively evenly spread within this range.



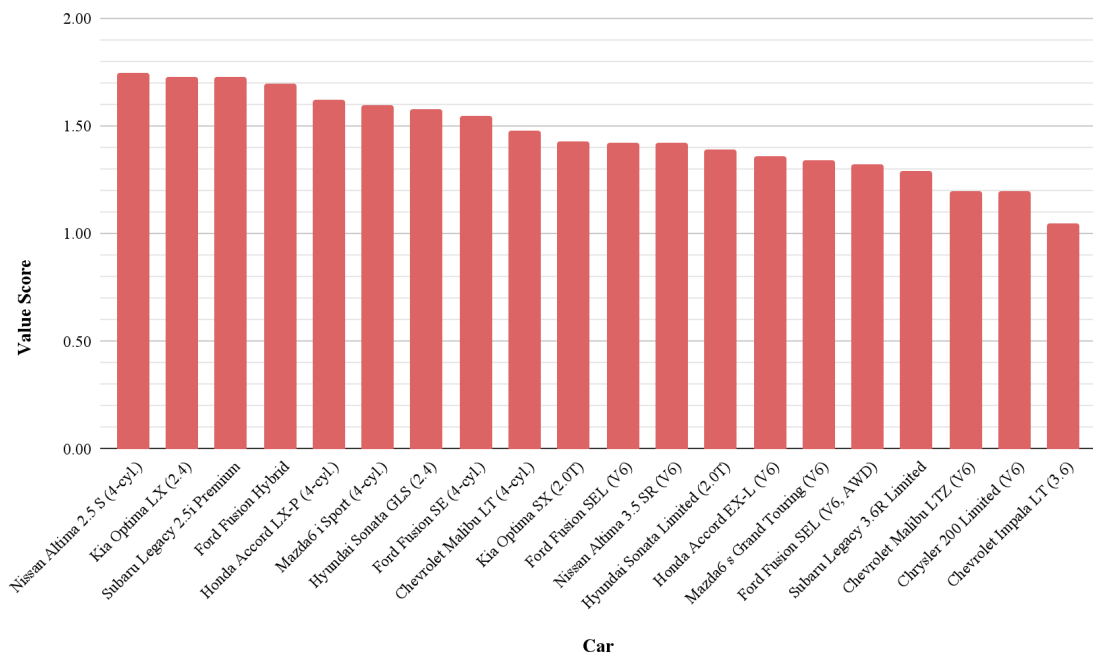
The histogram of cost per mile reveals how long-term ownership costs vary across the family sedans. Most vehicles cluster tightly around \$0.60–\$0.70 per mile, indicating that operating costs among family sedans are relatively similar. A few sedans fall toward the higher end of the distribution, which may signal less fuel-efficient or more maintenance-intensive models.

Road-Test Score



The road-test score histogram shows how performance ratings differ among the family sedans. The distribution demonstrates a wide spread, from lower-scoring sedans in the low 50s to top-performing models scoring above 90. This variation highlights meaningful differences in handling, comfort, and driving quality within the family sedan category.

Value Score



The value-score histogram indicates how Consumer Reports rates overall value among the family sedans. Most vehicles cluster around value scores of about 1.30 to 1.60, indicating that many sedans offer similar overall value. Only a few models score noticeably higher or lower, which emphasizes which family sedans stand out as exceptional or poor value relative to the group.

Value Score Given the Price of the Car

To develop an estimated regression equation that could be used to predict the value score given the price of the car, we must use linear regression analysis. For each regression model, we must find the estimated simple linear regression equation, expressed as:

$$\hat{y}_i = b_o + b_1 x_i$$

Where:

$\hat{y}_i$ = Estimated value of y for a given x value

b_o = Y-intercept of the estimated regression line

b_1 = Slope of the estimated regression line

x = Price (\$)

For the price of the car, the regression model that we must compute is:

$$\widehat{Value\ Score} = b_o + b_1(Price)$$

To calculate b_o and b_1 we must use the data given from the value score table and calculate each variable. We can compute the slope, b_1 , using the least squares method:

$$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

To calculate the slope, I created Excel columns for each component of the least-squares formula and then used Excel's SUM function to compute the required totals. Dividing the sum of the products by the sum of the squared deviations provided the value of b_1 :

Car	Price (\$)	Value Score	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
Nissan Altima 2.5 S (4-cyl.)	23,970.00	1.75	-2916.2	0.292	-851.5304	8504222.44
Kia Optima LX (2.4)	21,885.00	1.73	-5001.2	0.272	-1360.3264	25012001.44
Subaru Legacy 2.5i Premium	23,830.00	1.73	-3056.2	0.272	-831.2864	9340358.44
Ford Fusion Hybrid	32,360.00	1.70	5473.8	0.242	1324.6596	29962486.44
Honda Accord LX-P (4-cyl.)	23,730.00	1.62	-3156.2	0.162	-511.3044	9961598.44
Mazda6 i Sport (4-cyl.)	22,035.00	1.60	-4851.2	0.142	-688.8704	23534141.44
Hyundai Sonata GLS (2.4)	21,800.00	1.58	-5086.2	0.122	-620.5164	25869430.44
Ford Fusion SE (4-cyl.)	23,625.00	1.55	-3261.2	0.092	-300.0304	10635425.44
Chevrolet Malibu LT (4-cyl.)	24,115.00	1.48	-2771.2	0.022	-60.9664	7679549.44
Kia Optima SX (2.0T)	29,050.00	1.43	2163.8	-0.028	-60.5864	4682030.44
Ford Fusion SEL (V6)	28,400.00	1.42	1513.8	-0.038	-57.5244	2291590.44
Nissan Altima 3.5 SR (V6)	30,335.00	1.42	3448.8	-0.038	-131.0544	11894221.44
Hyundai Sonata Limited (2.0T)	28,090.00	1.39	1203.8	-0.068	-81.8584	1449134.44
Honda Accord EX-L (V6)	28,695.00	1.36	1808.8	-0.098	-177.2624	3271757.44
Mazda6 s Grand Touring (V6)	30,790.00	1.34	3903.8	-0.118	-460.6484	15239654.44
Ford Fusion SEL (V6, AWD)	30,055.00	1.32	3168.8	-0.138	-437.2944	10041293.44
Subaru Legacy 3.6R Limited	30,094.00	1.29	3207.8	-0.168	-538.9104	10289980.84
Chevrolet Malibu LTZ (V6)	28,045.00	1.20	1158.8	-0.258	-298.9704	1342817.44
Chrysler 200 Limited (V6)	27,825.00	1.20	938.8	-0.258	-242.2104	881345.44
Chevrolet Impala LT (3.6)	28,995.00	1.05	2108.8	-0.408	-860.3904	4447037.44
Average:	26,886.20	1.46			Sum: -7246.882	216330077.20
					b₁:	-0.000033

After finding b_1 , we can use the answer to calculate b_o using the least squares method:

$$b_o = \bar{y} - b_1 \bar{x}$$

$$b_o = 1.46 - (-0.000033)(26,886.20) = 2.358666$$

After finding b_o , we can develop the estimated regression equation by plugging in both slopes into the equation:

$$\widehat{Value\ Score} = 2.358666 - 0.000033(Price)$$

Value Score Given the Five-Year Owner Costs (Cost/Mile).

Using the same least-squares regression procedure described in Question 2, I estimated a new model with Cost/Mile as the predictor variable. The regression equation for predicting Value Score from Cost/Mile is:

$$\widehat{Value\ Score} = b_o + b_1(Cost/Mile)$$

Car	Cost/Mile	Value Score	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
Nissan Altima 2.5 S (4-cyl.)	0.59	1.75	-0.052	0.292	-0.015184	0.002704
Kia Optima LX (2.4)	0.58	1.73	-0.062	0.272	-0.016864	0.003844
Subaru Legacy 2.5i Premium	0.59	1.73	-0.052	0.272	-0.014144	0.002704
Ford Fusion Hybrid	0.63	1.70	-0.012	0.242	-0.002904	0.000144
Honda Accord LX-P (4-cyl.)	0.56	1.62	-0.082	0.162	-0.013284	0.006724
Mazda6 i Sport (4-cyl.)	0.58	1.60	-0.062	0.142	-0.008804	0.003844
Hyundai Sonata GLS (2.4)	0.56	1.58	-0.082	0.122	-0.010004	0.006724
Ford Fusion SE (4-cyl.)	0.57	1.55	-0.072	0.092	-0.006624	0.005184
Chevrolet Malibu LT (4-cyl.)	0.57	1.48	-0.072	0.022	-0.001584	0.005184
Kia Optima SX (2.0T)	0.72	1.43	0.078	-0.028	-0.002184	0.006084
Ford Fusion SEL (V6)	0.67	1.42	0.028	-0.038	-0.001064	0.000784
Nissan Altima 3.5 SR (V6)	0.69	1.42	0.048	-0.038	-0.001824	0.002304
Hyundai Sonata Limited (2.0T)	0.66	1.39	0.018	-0.068	-0.001224	0.000324
Honda Accord EX-L (V6)	0.67	1.36	0.028	-0.098	-0.002744	0.000784
Mazda6 s Grand Touring (V6)	0.74	1.34	0.098	-0.118	-0.011564	0.009604
Ford Fusion SEL (V6, AWD)	0.71	1.32	0.068	-0.138	-0.009384	0.004624
Subaru Legacy 3.6R Limited	0.71	1.29	0.068	-0.168	-0.011424	0.004624
Chevrolet Malibu LTZ (V6)	0.67	1.20	0.028	-0.258	-0.007224	0.000784
Chrysler 200 Limited (V6)	0.70	1.20	0.058	-0.258	-0.014964	0.003364
Chevrolet Impala LT (3.6)	0.67	1.05	0.028	-0.408	-0.011424	0.000784
Average:	0.64	1.46		Sum:	-0.164420	0.071120
						$b_1:$ -2.311867

$$b_o = 1.46 - (-2.311867)(0.64) = 2.942219$$

After finding b_o , we can develop the estimated regression equation by plugging in both slopes into the equation:

$$\widehat{Value\ Score} = 2.942219 - 2.311867(Cost/Mile)$$

Value Score Given the Road-Test Score

Applying the same regression steps as before, this model uses Road-Test Score as the independent variable.

$$\widehat{Value\ Score} = b_o + b_1(\text{Road-Test Score})$$

Car	Road-Test Score	Value Score	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
Nissan Altima 2.5 S (4-cyl.)	91	1.75	10.55	0.292	3.0806	111.3025
Kia Optima LX (2.4)	81	1.73	0.55	0.272	0.1496	0.3025
Subaru Legacy 2.5i Premium	83	1.73	2.55	0.272	0.6936	6.5025
Ford Fusion Hybrid	84	1.70	3.55	0.242	0.8591	12.6025
Honda Accord LX-P (4-cyl.)	80	1.62	-0.45	0.162	-0.0729	0.2025
Mazda6 i Sport (4-cyl.)	73	1.60	-7.45	0.142	-1.0579	55.5025
Hyundai Sonata GLS (2.4)	89	1.58	8.55	0.122	1.0431	73.1025
Ford Fusion SE (4-cyl.)	76	1.55	-4.45	0.092	-0.4094	19.8025
Chevrolet Malibu LT (4-cyl.)	74	1.48	-6.45	0.022	-0.1419	41.6025
Kia Optima SX (2.0T)	84	1.43	3.55	-0.028	-0.0994	12.6025
Ford Fusion SEL (V6)	80	1.42	-0.45	-0.038	0.0171	0.2025
Nissan Altima 3.5 SR (V6)	93	1.42	12.55	-0.038	-0.4769	157.5025
Hyundai Sonata Limited (2.0T)	89	1.39	8.55	-0.068	-0.5814	73.1025
Honda Accord EX-L (V6)	90	1.36	9.55	-0.098	-0.9359	91.2025
Mazda6 s Grand Touring (V6)	81	1.34	0.55	-0.118	-0.0649	0.3025
Ford Fusion SEL (V6, AWD)	75	1.32	-5.45	-0.138	0.7521	29.7025
Subaru Legacy 3.6R Limited	88	1.29	7.55	-0.168	-1.2684	57.0025
Chevrolet Malibu LTZ (V6)	83	1.20	2.55	-0.258	-0.6579	6.5025
Chrysler 200 Limited (V6)	52	1.20	-28.45	-0.258	7.3401	809.4025
Chevrolet Impala LT (3.6)	63	1.05	-17.45	-0.408	7.1196	304.5025
Average:	80.45	1.46		Sum:	15.2880	1,862.95
				b₁:	0.008206	

$$b_o = 1.46 - (0.008206)(80.45) = 0.797800$$

After finding b_o , we can develop the estimated regression equation by plugging in both slopes into the equation:

$$\widehat{Value\ Score} = 0.797800 + 0.008206(\text{Road-Test Score})$$

Value Score Given the Predicted Reliability

Following the same regression method outlined in Question 2, I computed a model using Predicted Reliability as the predictor.

$$\widehat{Value\ Score} = b_o + b_1(\text{Predicted Reliability})$$

Car	Predicted Reliability	Value Score	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
Nissan Altima 2.5 S (4-cyl.)	4	1.75	0.25	0.292	0.0730	0.0625
Kia Optima LX (2.4)	4	1.73	0.25	0.272	0.0680	0.0625
Subaru Legacy 2.5i Premium	4	1.73	0.25	0.272	0.0680	0.0625
Ford Fusion Hybrid	5	1.70	1.25	0.242	0.3025	1.5625
Honda Accord LX-P (4-cyl.)	4	1.62	0.25	0.162	0.0405	0.0625
Mazda6 i Sport (4-cyl.)	4	1.60	0.25	0.142	0.0355	0.0625
Hyundai Sonata GLS (2.4)	3	1.58	-0.75	0.122	-0.0915	0.5625
Ford Fusion SE (4-cyl.)	4	1.55	0.25	0.092	0.0230	0.0625
Chevrolet Malibu LT (4-cyl.)	3	1.48	-0.75	0.022	-0.0165	0.5625
Kia Optima SX (2.0T)	4	1.43	0.25	-0.028	-0.0070	0.0625
Ford Fusion SEL (V6)	4	1.42	0.25	-0.038	-0.0095	0.0625
Nissan Altima 3.5 SR (V6)	4	1.42	0.25	-0.038	-0.0095	0.0625
Hyundai Sonata Limited (2.0T)	3	1.39	-0.75	-0.068	0.0510	0.5625
Honda Accord EX-L (V6)	3	1.36	-0.75	-0.098	0.0735	0.5625
Mazda6 s Grand Touring (V6)	4	1.34	0.25	-0.118	-0.0295	0.0625
Ford Fusion SEL (V6, AWD)	4	1.32	0.25	-0.138	-0.0345	0.0625
Subaru Legacy 3.6R Limited	3	1.29	-0.75	-0.168	0.1260	0.5625
Chevrolet Malibu LTZ (V6)	3	1.20	-0.75	-0.258	0.1935	0.5625
Chrysler 200 Limited (V6)	5	1.20	1.25	-0.258	-0.3225	1.5625
Chevrolet Impala LT (3.6)	3	1.05	-0.75	-0.408	0.3060	0.5625
Average:	3.75	1.46		Sum:	0.8400	7.7500

$b_1:$ 0.108387

$$b_o = 1.46 - (0.108387)(3.75) = 1.051548$$

After finding b_o , we can develop the estimated regression equation by plugging in both slopes into the equation:

$$\widehat{Value\ Score} = 1.051548 + 0.108387(\text{Predicted Reliability})$$

Conclusion

Overall, the analysis illustrates that several factors can influence a car's value score, and each regression model helps explain a different part of that relationship. Higher prices tend to be associated with slightly lower value scores, while higher operating costs clearly reduce value by increasing long-term expenses. In contrast, cars with stronger road-test performance and higher predicted reliability generally earn higher value scores, since they offer better quality and fewer expected problems over time. By comparing all four models, it becomes clear that value is strongest when a car is affordable to own, performs well, and is expected to remain dependable. These results support the idea that the best-value cars are not always the cheapest, but those that balance cost, performance, and reliability consistently.